

FIN-220 Hands-On Machine Learning
HW 5
Due on November 17, 2025

1. What is the fundamental idea behind support vector machines?
2. What is a support vector?
3. Why is it important to scale the inputs when using SVMs?
4. Can an SVM classifier output a confidence score when it classifies an instance? What about a probability?
5. Say you've trained an SVM classifier with an RBF kernel, but it seems to underfit the training set. Should you increase or decrease γ ? What about C ?
6. What does it mean for a model to be ϵ -insensitive?
7. What is the point of using the kernel trick?